

PEM 2021 Year 11 Mathematics Standard Yearly Examination

SUGGESTED SOLUTIONS

Section 1 Multiple Choice Answer Key

Question	Answer	Outcomes Assessed	Targeted Performance Bands
1	<i>B</i>	<i>MS11–3</i>	2–3
2	<i>D</i>	<i>MS11–7</i>	4–5
3	<i>A</i>	<i>MS11–6</i>	2–3
4	<i>C</i>	<i>MS11–5</i>	3–4
5	<i>D</i>	<i>MS11–3</i>	4–5
6	<i>D</i>	<i>MS11–3</i>	4–6
7	<i>B</i>	<i>MS11–1</i>	2–3
8	<i>D</i>	<i>MS11–5</i>	4–5
9	<i>B</i>	<i>MS11–4</i>	3–4
10	<i>C</i>	<i>MS11–6</i>	4–5
11	<i>A</i>	<i>MS11–6</i>	2–3
12	<i>A</i>	<i>MS11–8</i>	3–5

Section II (Total 68 marks)**Question 13 A1**

Outcomes Assessed: MS11– 1

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	2
Makes progress towards the solution	1

Sample Answer:

$$7 - 2x = 3x - 3$$

$$\therefore 10 = 5x$$

$$\therefore x = 2$$

Question 14 (a) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	2
Makes progress towards the solution by finding the surface area of some of the walls	1

Sample Answer:

Surface area of 5 external walls of fish tank is

$$2 \times (108 \times 36 + 30 \times 36) + 108 \times 30 = 13176 \text{ cm}^2$$

Question 14 (b) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

$13176 \times \$0.12 = \1581.12 , i.e., it will cost approximately \$1581 to replace the glass.

Question 15 (a) A2

Outcomes Assessed: MS11– 6

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

$70 + 4 \times 30 = 190$. He pays \$190 to hire the boat.

Question 15(b)

Outcomes Assessed: MS11– 6

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

$\$190 \div 10 = \19

Each person pays \$19.

Question 15 (c)

Outcomes Assessed: MS11– 2

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct formula	1

Sample Answer:

$$C = 70 + 30h$$

Question 16 M1.1

Outcomes Assessed: MS11– 3

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$52.79 \times (2.3 \times 10^2) = 12141.7$$

This would be 12000 to 2 significant figures.

Question 17 A1

Outcomes Assessed: MS11– 6

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct substitution into formula	1
Provides correct solution	2

Sample Answer:

$$D = \frac{30 \times 400}{70} = 171.4 \approx 171 \text{ mg}$$

Question 18 (a) A2

Outcomes Assessed: MS11– 1

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

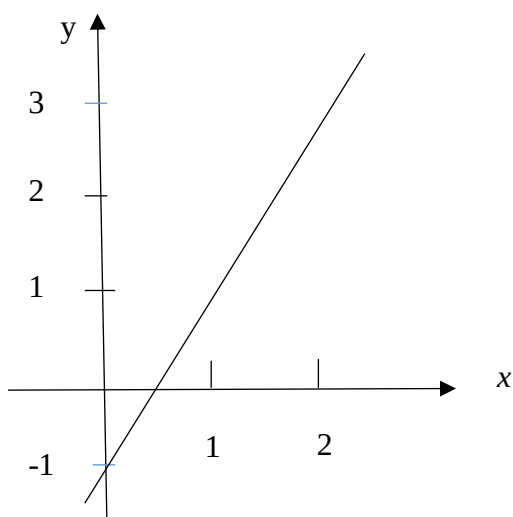
Gradient is 2 and y-intercept is -1

Question 18 (b) A2

Outcomes Assessed: MS11– 3

Targeted Performance Band: 3-4

Criteria	Marks
Draws the correct graph clearly labelled	2
Shows the y-intercept or gradient	1

Sample Answer:**Question 18 (c)**

Outcomes Assessed: MS11– 1

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$y = 0$$

$$2x - 1 = 0$$

$$2x = 1$$

$$x = \frac{1}{2}$$

Question 19 (a) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Makes progress towards the solution by finding correct angle or using correct formula	1

Sample Answer:

$$360^\circ - 45^\circ = 315^\circ$$

$$\frac{315^\circ}{360^\circ} \times \pi \times 1^2 = 2.75 \text{ cm}^2$$

Question 19 (b) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Provides some relevant calculation	1

Sample Answer:

$$\text{Area of sheet is } 7 \times 3 = 21 \text{ cm}^2$$

$$21 - 3 \times 2.75 = 12.75 \text{ cm}^2$$

Question 20 (a) F1.1

Outcomes Assessed: MS11– 10

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Makes progress towards the solution by finding difference in size or percent.	1

*Sample Answer:*Difference in size of apartment $80 - 77 = 3 \text{ m}^2$

$$\frac{3}{80} \times \frac{100}{1} = 3.75\%$$

Yes, the apartment is within the bounds of the contract.

Question 20 (b) A1

Outcomes Assessed: MS11– 6

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	2
Makes progress towards the solution by correct substitution into formula	1

Sample Answer:

Change the subject of the equation to S

$$S = SM \div 9.290304$$

$$S = 77 \div 9.290304$$

$$\approx 8.29 \text{ squares}$$

Question 21(a) S2

Outcomes Assessed: MS11-8

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$P(\text{correct guess 1st number}) = \frac{1}{10}$$

Question 21(b)

Outcomes Assessed: MS11-8

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$P(\text{correct guess 1st number and correct guess 2nd number}) = \frac{1}{10} \times \frac{1}{10} = \frac{1}{100}$$

Question 21 (c)

Outcomes Assessed: MS11-8

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Makes progress towards the solution	1

Sample Answer:

$$P(\text{correct guess 1st digit only}) = \frac{1}{10} \times \frac{9}{10} = \frac{9}{100}$$

$$P(\text{correct guess 2nd digit only}) = \frac{9}{10} \times \frac{1}{10} = \frac{9}{100}$$

$$P(\text{correct guess one digit only}) = \frac{9}{100} + \frac{9}{100} = \frac{18}{100}$$

Question 22 M1.3

Outcomes Assessed: MS11-3

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	3
Makes progress towards answer	2
Provides one correct calculation	1

Sample Answer:

Energy used to run the dishwasher per week is $1800 \times 2.5 \times 6 = 27000Wh$

$27000Wh = 27 kWh$

Cost per quarter is $27 \times 12 \times 12.199cents = \39.52

Question 23 (a) M1.2

Outcomes Assessed: MS11-4

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Shows use of trapezoidal rule	1

Sample Answer:

$$\text{Total Area} = \frac{4}{2}(0 + 3.7) + \frac{4}{2}(3.7 + 3.8) + \frac{4}{2}(3.8 + 0) = 30 \text{ m}^2$$

Question 23 (b)

Outcomes Assessed: MS11-4

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution in L	2
Calculates Volume in m ³	1

Sample Answer:

$$\text{Volume} = 30 \times 1.2 = 36 \text{ m}^3 = 36000 \text{ L}$$

since $1 \text{ m}^3 = 1000 \text{ L}$

Question 23 (c)

Outcomes Assessed: MS11-4

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$36000 \div 1000 = 36$$

$$40 \times 36 = 1440 \text{ mL} = 1.44 \text{ L}$$

Question 24 (a) S1.1

Outcomes Assessed: MS11-10

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct description	1

Sample Answer:

Positively skewed

Question 24 (b)

Outcomes Assessed: MS11-7

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

One or Unimodal

Question 24 (c)

Outcomes Assessed: MS11-7

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$85 - 0 = 85$$

Question 25 (a) M2

Outcomes Assessed: MS11-3

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution for day and time	2
Correct day, date or time	1

Sample Answer:

9pm + 17 hours is 2pm Thursday 8th of May

Question 25 (b) M2

Outcomes Assessed: MS11-3

Targeted Performance Band: 4-6

Criteria	Marks
Provides correct solution for day and time	2
Correct day, date or time	1

Sample Answer:

If Perth is 8 hours ahead of London

2pm Thursday is 10pm Thursday 8th of May

Question 26 (a) F1.2

Outcomes Assessed: MS11-5

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Makes progress towards answer	1

Sample Answer:

Yearly superannuation is $9.5\% \times \$92500 = \8787.50

Quarterly superannuation is $\frac{1}{4} \times \$8787.50 = \2196.88

Question 26 (b) F1.2

Outcomes Assessed: MS11-5

Targeted Performance Band: 4-6

Criteria	Marks
Provides correct solution	2
Makes progress towards correct solution	1

Sample Answer:

Taxable income is $\$92500 - \$1200 = \$91300$

Tax is $5092 + 32.5\% \times (\$91300 - \$45000) = \$20139.50$

Question 26 (c) F1.2

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

Medicare levy is $2\% \text{ of } \$91300 = \1826

Question 27 (a) F1.2

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

Upfront Commission is $0.65\% \times \$540000 = \3510

Question 27 (b) F1.2

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

Trail Commission is $0.15\% \times \$528154 = \792.23

Question 27 (c) F1.2

Outcomes Assessed: MS11-5

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Makes progress towards the solution by adding all the loans together.	1

Sample Answer:

$0.65\% \times (250000 + 625000 + 120000) = \6467.5

Student should recognise that the term is not needed for the calculation.

Question 28 (a) S1.2

Outcomes Assessed: MS11-7

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	3
Makes progress towards the solution 3-4 values	2
Provides some relevant information 1-2 values	1

Sample Answer:

Minimum: 1

Lower Quartile: 4

Median: 9

Upper Quartile: 18

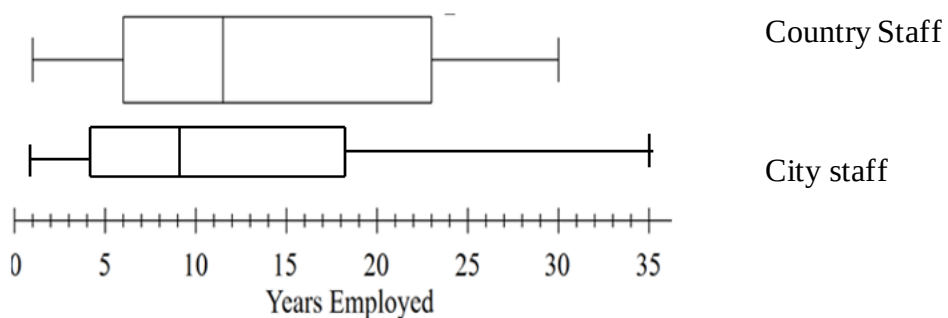
Maximum: 35

Question 28 (b) S1.2

Outcomes Assessed: MS11-11

Targeted Performance Band: 3-5

Criteria	Marks
Provides correct solution	2
Makes some progress towards the solution	1

Sample Answer:

Question 28 (c) S1.2

Outcomes Assessed: MS11-10

Targeted Performance Band: 3-5

Criteria	Marks
Provides correct solution	2
Makes some progress towards the solution	1

Sample Answer:

Similarities:

- Both departments are positively skewed.
- Neither group has any outliers.

Differences:

$IQR_{country} = 17 > IQR_{city} = 14$, i.e. The Country staff have the greater spread.

City staff have a larger range.

Question 29 (a) S1.1

Outcomes Assessed: MS11-2

Targeted Performance Band:

Criteria	Marks
Provides correct solution	1

Sample Answer:

Total participants are $19 + 11 + 7 + 2 + 1 = 40$

Question 29 (b) S1.1

Outcomes Assessed: MS11-10

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	2
Gives one suggestion	1

Sample Answer: He could make his presentation shorter and more engaging.

Question 30 (a) F1.1

Outcomes Assessed: MS11-5

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$3\% \times \$12000 = \$360$$

Question 30 (b) F1.1

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	2
Makes some progress towards the solution	1

Sample Answer:

Age	Amount in account
12	12000
13	12360
14	$12360 + 360 = 12720$
15	$12720 + 360 = 13080$
16	$13080 + 360 = 13440$
17	$13440 + 360 = 13800$
18	$13800 + 360 = 14160$

Kate will have \$14160 in her account at age 18.

Question 30 (c) F1.3

i) Outcomes Assessed: MS11-5, MS11-10

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	2
Makes some progress towards the solution	1

Sample Answer:

$$\$10000 \div 80 = 125 \text{ weeks}$$

ii) Outcomes assessed: MS11-3

Targeted Performance Band: 4-5

Provides correct solution	2
Makes some progress towards the solution	1

Sample Answer:

$$\frac{\$490}{40000} = \$0.01225 \text{ per kilometre}$$